

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 417 — Past Questions & Materials
by Renance

Course: PHY 417
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

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Paper 1 of 2 — PHY 417 Electronics

Level 400 • Second Semester • PastQuestion

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1a.



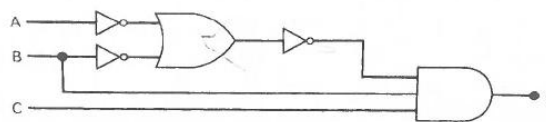
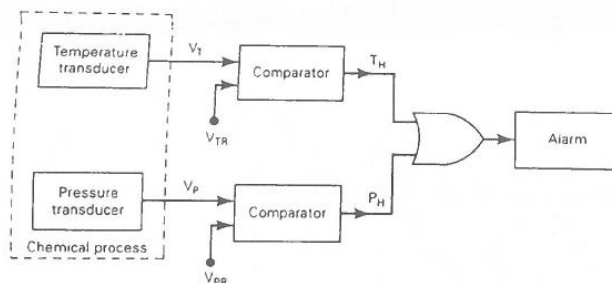
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF PHYSICS
SECOND SEMESTER EXAMINATION.

Course Code: PHY 417
Course Title: Electronics
Course Unit: 3
Session: 2023/2024

Time: 2hrs 30mins

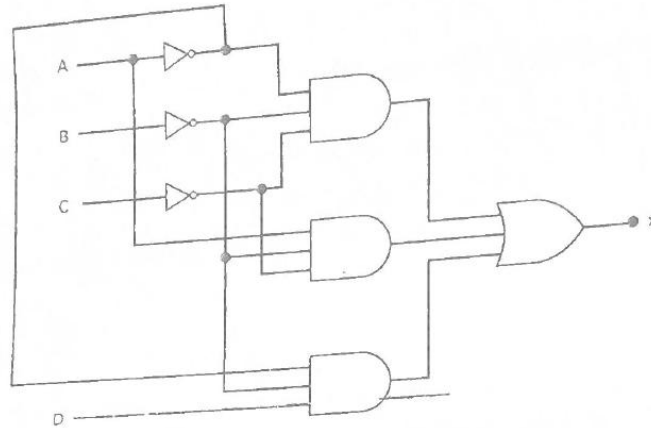
4

- 1a. How does analogue representation differ from digital representation?
- 1b. What are the main limitations to the use of digital techniques?
- 1c. What are the advantages of digital techniques and why is the electronic world rushing into the digital revolution?
- 1d. List the hexadecimal numbers in sequence from 280 to 2A0. *Write 280 to 299, 2A0*
- 1e. What is ASCII, what is its importance? *ASCII is an integrated circuit is used in str operation and information.*
- 2a. Modify the circuit below so that the alarm will be activated only when the pressure and the temperature exceed their maximum limits at the same time.



- 2b. Convert the circuit above to one using only NOR gates. Then write the expression for the new circuit. Simplify it using DeMorgan's theorems and compare it with the expression for the original circuit.

- 3a. Write the Boolean expression for output X in the diagram below. Determine the value of X for all possible input conditions and list the values in a truth table.



- 3b. For each of the following expressions, construct the corresponding logic circuit using AND and OR gates and inverters.

- $X = \overline{AB(C + D)}$
- $z = (A + B)(\bar{A} + \bar{B})$

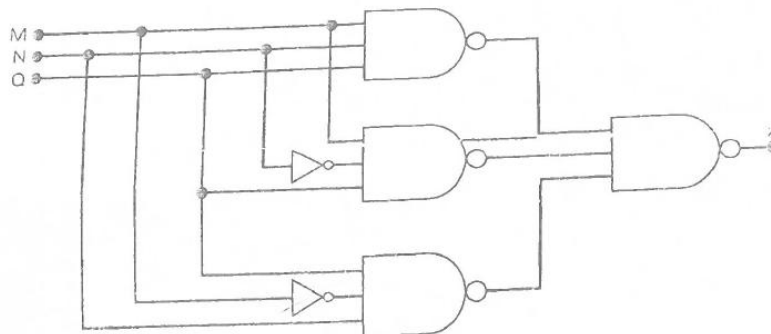
- 4a. Design the logic circuit corresponding to the truth table shown below.

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

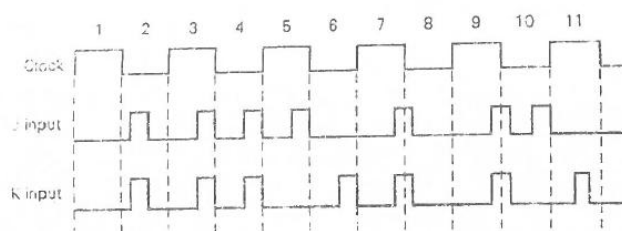
- 4b. The following numbers are in BCD Convert them to decimal;

- 1001011101010010
- 01110111101110101

- 4c. Simplify the circuit below using Boolean algebra.



- 5a. Distinguish between Asynchronous and synchronous counters.
- 5b. With the aid of diagrams, explain how flip-flops are used to implement the following processes:
- Data storage and transfer
 - Frequency division
 - Counting
 - Debouncing a mechanical switch



- 5a. The waveform shown in the figure above are to be applied to two different Flip-flops;
- Positive-edge triggered J-K
 - Negative-edge triggered J-K

Draw the Q waveform response for each of these FFs assuming that $Q = 0$ initially.

- 6b. The owners Manual for a small microcomputer states that the computer has useable memory locations at the following hex addresses; 0200 through 03FF and 4000 through 7FD0. What is the total number of available memory locations?

Paper 2 of 2 — Electronics PHY 417 PQ

Level 400 • First Semester • PastQuestion

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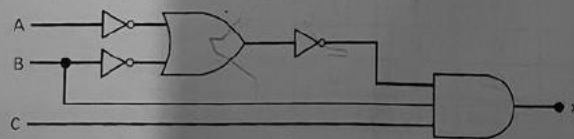
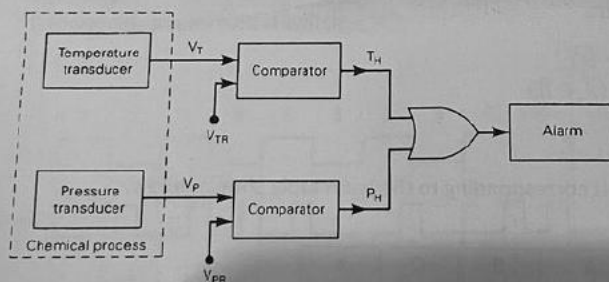
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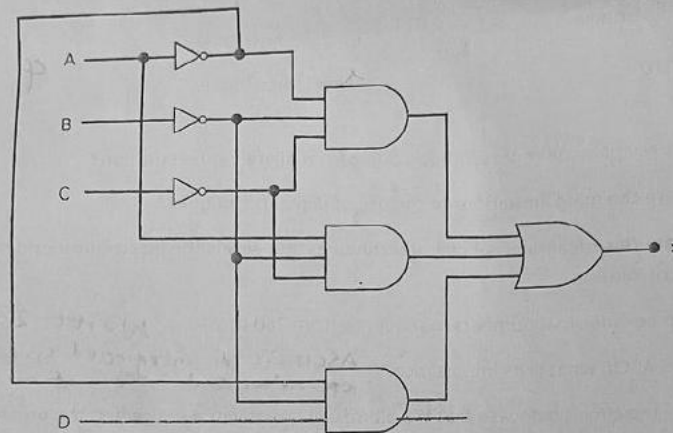
4

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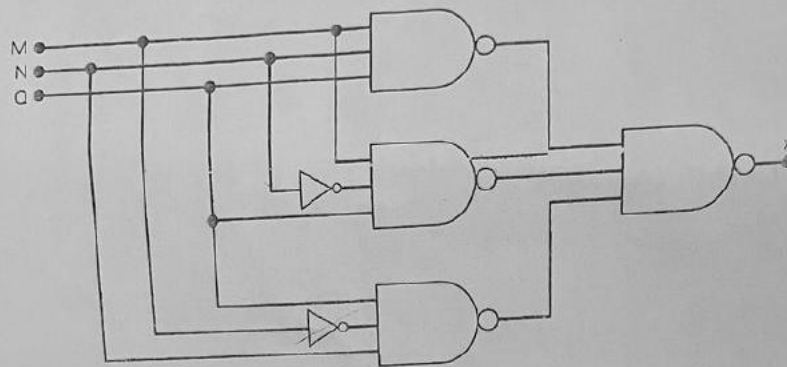
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1	0	1	0
1	1	0	0
1	1	1	1

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- 10010111010010
- 01110111101110101

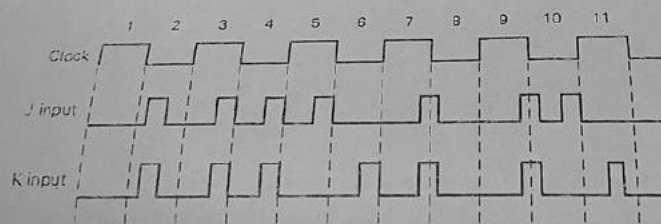
4c. Simplify the circuit below using Boolean algebra.



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5b. With the aid of diagrams, explain how flip-flops are used to implement the following processes.

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- iv. Debouncing a mechanical switch



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